ENVIRONMENT & BIODIVERSITY

Exam-Ready Notes | Unit I

Environmental Studies | RVCE

1. ENVIRONMENT — Definitions, Scope & Importance

1.1 Key Definitions

Environment	The whole physical and biological system in which man and other organisms live. Involves every issue that affects living organisms.

Environmental Science	A multidisciplinary study of the functioning of nature and interconnections between various things in nature. Components: biology, geology, chemistry, physics, engineering, sociology, health, economics.

1.2 Scope

- Natural landscapes (forest, river, desert, rocks, minerals, soil) are transformed by humans into villages, towns, cities.
- Daily life is linked to surrounding environment — we need water, air, food.
- Most environmental resources (water, minerals, petroleum, wood) are continuously extracted and are NON-PRODUCIBLE — once exhausted, cannot be reproduced in original form.
- Technological advancement (fertilizers, pesticides, dams) leads to environmental degradation.

1.3 Importance

- Environmental resources play multifunctional roles and command market prices — scarce resources cost more (e.g., wood, water).
- Deforestation → increasing temperature, dry rivers, unavailability of fresh air → diseases.
- Misuse/waste of natural resources can be stopped only by spreading awareness.

2. NEED FOR PUBLIC AWARENESS

2.1 People in Environment

- Environmental movement covers: land, water, minerals, living organisms, atmosphere, climate, oceans, outer space.
- Notable environmentalists: Charles Darwin, Salim Ali, Indira Gandhi, S.P. Godrej, Madhav Gadgil, M.C. Mehta, Medha Patkar, Sundarlal Bahuguna.
- Public participation: pressure groups, watchdog/observer, advisory councils, reinforcing environmental laws.

2.2 NGOs & Media

- Key NGOs: Friends of Earth, Greenpeace, WWF (World Wide Fund for Nature).
- Media is a powerful but reactive instrument in raising public awareness.

2.3 Environmental Education — Scope & Importance

Scope	Importance
Creates awareness and sensitivity to environmental problems	Understand 'development without destruction'
Active participation in environment protection	Understand environmental hazards
Develop skills to identify and solve environmental problems	Demand laws for protecting environment
Understand the need to conserve natural resources	Relate quality of life with environment

3. ECOSYSTEMS — Structure, Function & Types

3.1 Definition

An ecosystem is a community of different species interacting with one another and with their environment, exchanging energy and matter. 'Eco' = environment + 'System' = interacting, inter-dependent complex.

> 🧠 ***MNEMONIC: BAPS: Biotic + Abiotic + Producers + System = Ecosystem***

3.2 Classification of Ecosystems

Type	Examples
Natural: Terrestrial	Forest, Grassland, Desert ecosystems
Natural: Aquatic — Fresh water	Running water (rivers, streams); Standing water (lake, pond)
Natural: Aquatic — Marine	Saltwater/ocean ecosystem
Artificial / Man-made	Cropland, gardens, aquarium

> 🧠 ***MNEMONIC: NATA: Natural → Aquatic/Terrestrial; Artificial → by man***

3.3 Characteristics of Ecosystems

- Structural and functional unit of ecology.
- Structure related to species diversity — more complex ecosystem = higher species diversity.
- Function related to energy flow and cycling of materials.
- Ecosystem matures from simple to complex (early stage: excess potential energy; mature stage: energy flow stabilizes).
- Alteration in environment = selective pressure → survival of fittest.
- Energy fixation in any ecosystem is limited — exceeding it causes serious effects.

3.4 Difference: Natural vs Artificial Ecosystems

Natural Ecosystem	Artificial / Man-made Ecosystem
Operates under natural conditions	Maintained artificially by man
No human interference	Energy added and manipulated through planning
Self-sustaining	Requires external inputs (fertilizers, irrigation)
Example: Forest, Pond, Desert	Example: Cropland, Garden, Aquarium

> ⚡ ***EXAM TIP: A common exam question is to differentiate natural vs artificial ecosystems — always include examples!***

4. STRUCTURE & COMPONENTS OF AN ECOSYSTEM

4.1 Abiotic (Non-Living) Components

Physical: Sunlight (photosynthesis), Water (essential for life), Temperature (survival), Soil (base + nutrients).

Chemical: Carbohydrates, proteins, lipids, nitrogen, phosphorous, potassium, oxygen.

4.2 Biotic (Living) Components

Component	Description	Examples
Producers / Autotrophs	Use sunlight + soil nutrients for photosynthesis; 1st trophic level	All green plants, algae
Consumers / Heterotrophs — Herbivores (Primary)	Eat producers directly	Man, elephant, rabbit
Consumers — Carnivores (Secondary)	Eat primary consumers	Tiger, lion
Consumers — Omnivores	Eat both producers and animals	Fox, frog
Decomposers / Detrivores	Recycle nutrients; break down dead matter into simple compounds	Bacteria, fungi

> 🧠 ***MNEMONIC: PAC-D: Producers → All Consumers → Decomposers (flow of energy in biotic components)***

5. ENERGY FLOW IN ECOSYSTEM

5.1 Basic Concept

The energy needed for ecosystems comes from the SUN. Solar energy is converted into chemical energy (carbohydrates + oxygen) via photosynthesis.

Photosynthesis equation: $CO_2 + 2H_2O \rightarrow$ (sunlight/chlorophyll) $\rightarrow CH_2O + O_2 + H_2O$

- A part of chemical energy is used by producers for their own growth.
- Remaining energy is transferred to consumers.
- Decomposers utilize the energy from consumers, producing inorganic nutrients, which are reused by producers.

5.2 Nutrient Cycle / Biogeochemical Cycle

Nutrients flow repeatedly between biotic and abiotic components — this is the nutrient cycle (or biogeochemical cycle).

Types of biogeochemical cycles: Hydrological (Water), Oxygen, Nitrogen, Carbon, Phosphorous.

5.3 Biogeochemical Cycles — Summary

Cycle	Key Process / Notes
Hydrological (Water)	Continuous movement of water between earth surface and atmosphere via evaporation and precipitation.
Carbon Cycle	CO_2 removed by photosynthesis; returned via respiration, combustion, volcanic eruptions.
Nitrogen Cycle	N_2 = 78% of atmosphere. Nitrification ($NH_3 \rightarrow$ Nitrates) by Nitrobacter, Nitrosomonas. Denitrification (Nitrates $\rightarrow N_2$) by Pseudomonas.
Phosphorous Cycle	Found in rocks/fossils. Excavated for fertilizers → runoff to oceans. Sea birds return phosphorus to land.

> 🧠 ***MNEMONIC: HCNP: 'Hot Climates Need Phosphorus' = Hydrological, Carbon, Nitrogen, Phosphorous cycles***

6. TYPES OF ECOSYSTEMS — Structure & Function

6.1 Forest Ecosystem

- Tall and dense trees supporting many animal species.

Types: Tropical rain forests, Tropical deciduous, Temperate deciduous, Tropical scrub forests, Temperate rain forests.

Abiotic: Climatic factors (temperature, rainfall, light), minerals.

Biotic — Producers: Trees, shrubs, ground vegetation.

Biotic — Consumers: Ants, flies, insects, mice, deer, snakes, birds, tiger, lion.

Biotic — Decomposers: Bacteria, fungi.

Characteristics: Warm climate with adequate rainfall; well-defined seasons; rich soil; slow growth; protects biodiversity.

6.2 Grassland Ecosystem

- Variety of grasses, herbs, insects. Grass is the major producer of biomass.

Types: Tropical, Temperate, Polar grassland.

Abiotic: CO_2, H_2O, nitrate, phosphates, sulphates (C, H, O, N, P, S).

Biotic — Producers: Grasses, forbs, shrubs.

Primary consumers: Cows, buffaloes, deer, sheep.

Secondary consumers: Snakes, lizards, birds.

Tertiary consumers: Hawks, eagles.

Decomposers: Fungi, bacteria.

Characteristics: Low and uneven rainfall; rich soil; largest biomass; moderate climate.

6.3 Desert Ecosystem

- Ecological succession of grassland — high temperature, less moisture, less vegetation.

Four major types: Hot & dry, Semiarid, Coastal, Cold desert.

Abiotic: Temperature, rainfall, sunlight, water.

Producers: Succulent plants (cacti), shrubs, bushes (waxy layer protects from sun).

Consumers: Mice, rabbits, reptiles, squirrels — extract water from seeds.

Characteristics: Strong winds; low annual rainfall; dry hot air; large temperature variation (hot day/cold night); no/rare vegetation; no soil.

6.4 Aquatic Ecosystems

Types: Fresh water (rivers, ponds, lakes, streams, wetlands) and Marine (ocean, estuary).

6.4a Pond Ecosystem

- Freshwater, stagnant (standing) water body — temporary (monsoon).
- Abiotic: Light, temperature, dissolved O_2, pH, phosphorous.
- Producers: Phytoplankton (algae, volvox), Microphytes (hydrilla, wolfia).
- Consumers: Zooplanktons (protozoa, small fish) → small carnivores (water beetles) → large fish.
- Decomposers: Fungi, bacteria, flagellates.
- Characteristics: Stagnant water, less nutrients, gets polluted easily, mostly temporary.

6.4b Lake Ecosystem

Type of Lake	Description
Oligotrophic	Low nutrient concentration
Eutrophic	Overnourished nutrients
Dystrophic	Low pH, brown water, acidic
Volcanic	Receives water from magma
Meromictic	Salt-rich

6.4c Ocean (Marine) Ecosystem

- Largest of all ecosystems; high salt and mineral concentration.
- Producers: Phytoplanktons (diatoms, unicellular algae), seaweeds.
- Primary consumers: Crustaceans, molluscs, fish. Secondary: Herring, mackerel. Tertiary: Cod, haddock.
- Characteristics: Large surface area; rich biodiversity; algae absorb CO_2 and provide oxygen.

7. BIODIVERSITY

7.1 Definition

Biodiversity (Biological diversity) = Variety and variability of living organisms in a given assemblage. Covers all life on earth. Described in terms of GENES, SPECIES, and ECOSYSTEMS.

7.2 Types of Biodiversity

Type	Definition	Example
Genetic Diversity	Variety within the same genes within individual species; based on variation between functional units of hereditary information	Thousands of rice varieties; each human being is unique
Species Diversity	Number of different species of living things within an area; members of one species don't breed freely with another	Tiger, lion, teakwood, human beings
Ecosystem Diversity	Variety of habitats, biotic communities and ecological processes in the biosphere; includes functional, community and landscape diversity	Tropical forest vs. Arctic tundra

> 🧠 ***MNEMONIC: GSE: 'Gene → Species → Ecosystem' — from smallest to largest scale of biodiversity***

7.3 Importance of Biodiversity

- Increases ecosystem productivity — each species has a specific role.
- Supports larger number of plant species → greater variety of crops.
- Protects freshwater resources.
- Promotes soil formation and protection.
- Provides nutrient storage and recycling.
- Aids in breaking down pollutants.
- Contributes to climate stability.
- Speeds recovery from natural disasters.
- Provides food, medicinal resources, pharmaceutical drugs.
- Offers environments for recreation and tourism.

> 🧠 ***MNEMONIC: FSSN-CADF-MT: Food, Soil, Support, Nutrient, Climate, Aid recovery, Drugs, Fresh water, Medicines, Tourism***

8. VALUES OF BIODIVERSITY

Value Type	Description	Examples
Consumptive Use	Direct utilization of species by modern society	Food (fruits, seafood), Medicinal plants (neem, eucalyptus), Fuel (firewood, coal, petroleum)

Productive Use	Commercial products synthesized from natural products	Silk, wool, leather, tusk (animals); wood, cotton, oil seed (plants)
Social Values	Biodiversity for social/cultural/religious aspects; ecotourism	Holy plants: Banyan, peepal, lotus; Holy animals: Cow, peacock, snake
Ethical Values	Every species has equal right to live; conservation of life	Mission of communities to preserve animal life; India's heritage of worshipping nature
Aesthetic Values	Beauty, wonder, joy, recreational pleasure	Wildlife tourism; ornamental plants; elephants/horses for ceremonies; neem/mango leaves in festivals
Option Values	Potential future uses not yet known	Rare medicinal plants that may cure chronic diseases

> 🧠 ***MNEMONIC: CPSEA-O: 'Can People See Every Aesthetic Option?' = Consumptive, Productive, Social, Ethical, Aesthetic, Option***

9. HOT-SPOTS OF BIODIVERSITY

9.1 Definition & Criteria

Hot-spots = Specific areas containing the richest and most threatened reservoirs of plant and animal species. Total: 25 hot-spots identified worldwide, covering about 1.4% of total land surface.

Criteria to qualify as a hot-spot:

- Richness of endemic species.
- Significant percentage of specified species must be present.
- Must have lost MORE THAN 70% of its original habitat.
- Must be under threat.

9.2 Hot-spots in India (2 out of 25 worldwide)

Hot-spot	Key Facts
Eastern Himalayas	Comprises Nepal, Bhutan, northern India, Yunnan province (SW China). ~35,000 plant species in Himalayas; 30% endemic. Sikkim: 4,250 plant species, 60% endemic. Ultra-varied topography → high endemism.
Western Ghats	Extends ~1600 km along western coast (Tamil Nadu, Maharashtra, Karnataka, Kerala). Only 6.8% of original vegetation survives. Out of India's 49,219 plant species, 1,600 endemics (40%) are here. Common plants: Temstroemia Japonica, Rhododendron, Hypericum.

9.3 Reasons for Rich Biodiversity in Tropics

- More stable climate.
- Warm temperature + high humidity → favourable conditions.

- No single species can dominate → opportunity for many species to coexist.
- Higher rate of outcrossing among plants.

10. THREATS TO BIODIVERSITY

Major Threats

Threat	Key Details
Habitat Loss	Rapid human population growth; Deforestation; Destruction of wetlands; Overgrazing; Urban development; Dam building; Mining; Land slides; Poor agricultural practices; Industrial wastes.
Poaching of Wildlife	Hunting for fur, tusks, meat, thorns. Elephant (ivory), Alligators (boots), Tiger (skin/bones), Rhinos (horns), Deer (musk).
Man-Wildlife Conflicts	Human encroachment reduces forest size → animals attack human settlements. Monoculture plantations (teak, sal) create ecological imbalance.
Other Threats	Destruction of coastal areas; Filling up of wetlands; Commercial exploitation; Introduction of invasive species.

> ***⚡ EXAM TIP: The PYQ on threats frequently asks for habitat loss in detail. Always include the sub-factors list.***

Impact of Human Activities on Biodiversity Loss

- Deforestation for timber → monoculture plantations → ecological imbalance.
- Introduction of alien species disturbs balance of existing community.
- Man interacts with ecosystems for food, fuel, recreation, urban development, waste disposal — all directly or indirectly disturb species.
- Over-exploitation of natural resources → degradation and depletion.

11. CONSERVATION OF BIODIVERSITY

11.1 Advantages of Conservation

- Ecotourism → good source of income.
- Provides various medicinal plants.
- Provides life support system on earth.
- Maintains environmental balance.
- Various commercial aspects related to biodiversity.

11.2 Two Approaches

In-situ Conservation (On-site)	Ex-situ Conservation (Off-site)
Protection of species where they naturally exist	Conservation outside natural habitats, under human control
Protected areas: national parks, sanctuaries, forests, biosphere reserves, botanical gardens	Botanical gardens, zoos, aquariums, seed banks, gene banks, cryo-preservation
Tourism, poaching, shooting, grazing, tree cutting are strictly prohibited	Breeding under controlled conditions
Advantages: Preserves ecosystem; protects individual species; natural habitat maintained	Advantages: Assured food/water/shelter → longer life; special care for endangered species; modern breeding facilities
Disadvantages: Problems of encroachment, illegal activities, maintenance	Disadvantages: Expensive; species can't adapt to changing natural conditions; loss of freedom

> 🧠 ***MNEMONIC: In-situ = 'In the SITE' (natural habitat). Ex-situ = 'EXit the site' (away from habitat).***

11.3 Methods of In-situ Conservation

Method	Available in India	Notes
Biosphere Reserves	7 (listed; 12 total by 2000)	Human population constitutes an important component; used for education, research, recreation; 408 reserves globally in 94 countries.
National Parks	80	Area dedicated for conservation of wildlife along with its environment. Ex: Kaziranga (Rhino), Gir (Indian Lion), Bandipur (Elephant), Corbett (Tiger).
Wild-life Sanctuaries	420	Area reserved for conservation of animals only. Ex: Hazaribagh (Tiger/Leopard), Ghana Bird Sanctuary (300 bird species).
Botanical Gardens	120	Conservation of plant species.

11.4 Steps for Conservation of Biodiversity

- Biodiversity inventories and population surveys.

- Identifying and expanding protected areas.
- Conserving biodiversity in seed banks and gene banks.
- Controlling wildlife trade.
- Providing environmental education.
- Reviewing agricultural practices.
- Controlling urbanization.
- Geographic Information System (GIS) for planning and monitoring.
- Restoration of biodiversity.
- Population control.
- Implementing Environmental Protection Act (EPA).
- Involving NGOs.

11.5 National Biodiversity Act (2002)

India is party to the Convention on Biological Diversity (CBD) 1992. India enacted the Biological Diversity Act 2002. National Biodiversity Authority established on 1st October 2003 (headquarters: Chennai).

Main Functions: Lay down procedures for obtaining biological resources; advise government on threatened species; encourage state biodiversity boards; build databases; create awareness through media.

12. PRIMARY vs SECONDARY SUCCESSION

(This topic appears in syllabus as 'ecological succession' — commonly asked as a comparison)

Aspect	Primary Succession	Secondary Succession
Definition	Development of ecosystem from bare/lifeless substrate where no life existed before	Re-development of ecosystem in a place where a previous community was destroyed
Initiating Factor	Bare rock, newly formed volcanic island, sand dunes	Forest fire, flood, deforestation, agriculture abandonment
Starting Community	Pioneer species (e.g., lichens on rock)	Remnants of previous community; seeds and spores remain in soil
Rate of Development	Very slow (hundreds to thousands of years)	Faster (decades to centuries) — soil and nutrients already present
Species Diversity	Starts very low; gradually increases	Higher from the start — existing soil supports faster colonization
Soil	Absent initially; must be formed by pioneer species	Already present and relatively fertile
End Point	Climax community (stable, self-sustaining)	Return to climax or near-climax community

🧠 ***MNEMONIC: Primary = 'Pioneer on bare Planet'. Secondary = 'Second chance after disturbance'.***

13. ENVIRONMENTAL POLLUTION — Overview

Pollution = any substance introduced into the environment that adversely affects the usefulness of a resource. Can be solid, liquid, or gas.

Type of Pollutant	Description	Examples
Bio-degradable (Non-persistent)	Rapidly decomposed by natural processes	Domestic sewage, discarded vegetables
Slowly degradable (Persistent)	Decompose very slowly; remain long in environment	Plastics, pesticides
Non-degradable	Cannot be decomposed by natural processes	Lead, mercury, nuclear wastes

Classification of Pollution: Air pollution, Water pollution, Soil pollution, Marine pollution, Noise pollution, Thermal pollution, Nuclear hazards.

14. AIR POLLUTION

14.1 Definition & Causes

Air pollution = undesirable contamination of gas, smoke, dust, fume, mist, odour or chemical particulates in the atmosphere — injurious to human beings, plants and animals.

Causes: Industrialization, Urbanization, Vehicle emission, Deforestation, Population growth.

14.2 Primary vs Secondary Pollutants

Aspect	Primary Pollutants	Secondary Pollutants
Source	Emitted directly from natural events or human activities	Formed when primary pollutants react with each other or with air components
Examples	CO, CO_2, NO_x, SO_x, Hydrocarbons, Particulate matter (account for 90% of global air pollution)	Sulphuric acid (H_2SO_4), Nitric acid (HNO_3), Carbonic acid, Photochemical smog (ozone)
Natural sources	Dust storms, volcanoes	Formed in atmosphere by chemical reactions
Human sources	Vehicle exhaust, industrial waste	Reaction products of primary pollutants under sunlight

> 🧠 ***MNEMONIC: Primary = 'Produced directly'. Secondary = 'Son of Primary' (derived from reactions).***

14.3 Common Air Pollutants

Pollutant	Sources	Effects

Carbon Monoxide (CO)	Automobile exhaust, cigarettes, coal/biomass combustion, incomplete combustion	Headaches, dizziness, drowsiness; ~11% of heart failures; brain damage at high doses; displaces O_2 in blood
Sulphur Dioxide (SO_2)	Chemical industries, metal smelting, pulp/paper mills, oil refineries, coal combustion (coal = single largest source)	Reacts with moisture → irritating acid; triggers allergies, asthma; acid deposition (H_2SO_4); ozone depletion
Nitrogen Dioxide (NO_2)	Motor vehicle exhausts, gasoline, volcanoes, lightning	Irritates eyes; causes asthma, infection; poisonous to plants; forms HNO_3 (corrodes metals/stone)
Lead (Pb)	Combustion of leaded petrol (80–90%), paint, storage batteries	Mental retardation, digestion problems, cancer; harmful to wildlife; children especially sensitive
Particulate Matter (SPM)	Power plants, industrial processes, motor vehicles, domestic coal burning	Penetrates respiratory system; particles <4μm enter alveoli; respiratory distress, lung impairment, mortality
Photochemical Smog	Formed from reaction of NO_x + volatile organic compounds + sunlight	Eye, nose and throat irritation; chest discomfort; coughs, headache; modern industrialization problem

14.4 Control Measures for Air Pollution

Source Control

- Use unleaded petrol only.
- Use petroleum products with low sulphur content.
- Use public transport instead of private vehicles.
- Plant trees (absorb particulates, CO_2, and noise).
- Industries and waste disposal sites should be outside city (downwind).
- Use catalytic converters to control CO and hydrocarbon emissions.

Industrial Control

- Restrict emission rates to permissible levels.
- Mandatory air pollution control equipment in plant design.
- Continuous monitoring of emissions.
- Equipment: Scrubbers, Cyclones, Baghouses (bag filters), Electrostatic precipitators.
- Wet scrubber removes SO_2 emissions.

15. WATER POLLUTION

15.1 Definition

Any physical, biological or chemical change in water quality that adversely affects living organisms or makes water unsuitable for certain uses.

15.2 Types, Causes & Effects

Type of Pollution	Cause/Source	Effects
Infectious Agents	Human/animal wastes; bacteria, virus, protozoa	Amoebic dysentery, malaria, skin problems
Oxygen Demanding Wastes (BOD)	Sewage, animal feedlots, paper mills, food processing	High BOD depletes O_2 → kills aquatic life
Inorganic Chemicals	Surface runoff, industrial effluents; lead, arsenic, fluorides	Skin cancers, nervous system damage, corrosion
Organic Chemicals	Industrial effluents; oil, gasoline, pesticides, detergents	Cancer, liver/nervous system damage, harms aquatic life
Plant Nutrients (Eutrophication)	Sewage, agricultural runoff; nitrate, phosphate	Excessive algal growth → kills aquatic life; blue baby syndrome (nitrates in groundwater)
Sediment	Land erosion; soil, silt	Reduces photosynthesis; disrupts food web; carries pesticides/bacteria
Radioactive Materials	Nuclear power plants, weapons; uranium, thorium	Genetic mutations, birth defects, cancer
Thermal Pollution	Industrial water cooling processes	Aquatic organisms become vulnerable to diseases

15.3 Point vs Non-Point Source Pollution

Point Source	Non-Point Source
Single identifiable source	Sources are scattered or diffuse
Easy to monitor and regulate	Difficult to identify and regulate
Examples: Industrial discharge, factory smoke stack, municipal sewage	Examples: Farm runoff, construction sites, parking lots, agricultural logging, animal waste

15.4 Effects of Water Pollution

On humans: Amoebic dysentery, cholera, typhoid, skin cancers, nervous system damage, genetic mutations, hepatitis, malaria.

On plants/animals: Lower crop yields; harmful to aquatic/wild life; excess algae kills aquatic life; disrupts food chain.

15.5 Control Measures for Water Pollution

- Set up effluent treatment plants.
- Encourage water recycling.
- Treat industrial wastes before discharge.
- Educate public on water pollution.
- Strict enforcement of water pollution control act.

- Continuous monitoring of water quality.
- Develop economical water treatment methods.
- Protect rivers, lakes, and water reservoirs from pollution.

16. SOIL POLLUTION

16.1 Definition

Contamination of soil causing adverse effects on living organisms in it.

16.2 Causes of Soil Pollution

Cause	Details
Soil Erosion	Movement of topsoil by wind/flood; human activities: construction, overgrazing, farming, deforestation
Industrial Wastes	Discharge from chemical, fertilizer, pharmaceutical companies
Urban Wastes	Domestic and commercial wastes; plastic (non-degradable)
Agricultural Practices	Strong fertilizers, pesticides, inorganic chemicals for increasing yield
Biological Agents	Human/animal excreta → pathogenic bacteria → infections

16.3 Effects of Soil Pollution

- Toxic compounds affect plant growth and human life.
- Waterlogging and salinity make soil infertile.
- Hazardous chemicals enter food chain from soil → disrupt biochemical processes.
- Nervous disorders, gastrointestinal disorders, joint pain, respiratory problems in humans.

16.4 Control Measures for Soil Pollution

- Prevent/control soil erosion by proper tree plantation.
- Dump industrial and domestic wastes only after proper treatment.
- Prefer natural fertilizers over synthetic ones.
- Educate people about soil pollution.
- Strict enforcement of environment protection laws.
- Ban toxic and non-degradable materials.
- Recycling and reuse of industrial/domestic wastes.

16.5 Impacts of Modern Agriculture

Fertilizer Problems

Problem	Details

Micronutrient Imbalance	Excess use of macronutrient fertilizers (N, P, K) → imbalance of micronutrients → reduces soil productivity
Blue Baby Syndrome	Nitrates leach into groundwater → unacceptable for drinking → causes 'Methemoglobinemia' (Blue Baby Syndrome) → can lead to death
Eutrophication	Excess phosphorous in fertilizers carried by runoff → water bodies → excessive algal growth → kills aquatic life

Pesticide Problems

- Kill non-target organisms (beneficial insects).
- Create 'superpests' — resistant strains immune to pesticides.
- Bio-magnification — pesticides concentrate in food chain.
- Risk of cancer — act as carcinogens; suppress immune system.
- Contaminate groundwater via surface runoff.

17. NOISE POLLUTION

17.1 Definition & Overview

Unwanted, unpleasant sound causing discomfort to human beings. Measured in decibels (dB). Differs from other pollutions — leaves no residual in environment if stopped; effects felt only close to source.

17.2 Sources

Industrial: Steel industry, textile industry, power generation, oil refineries.
Transport: Road, rail, and air traffic.
Domestic: Mixer, washing machine, refrigerator, AC, vacuum cleaners, TV, radio.

17.3 Effects

- Neurological disorder, anxiety, mental distress.
- Heart attacks, pathological disorder.
- Deafness / impairment of hearing.
- Sleeplessness, psychiatric illness.
- Ultrasonic sound affects digestive, respiratory, cardiovascular systems and semicircular canals of inner ear.

17.4 Ambient Noise Levels

Zone	Day-time	Night-time
Silent zone	50 dB	40 dB
Residential zone	55 dB	45 dB
Commercial zone	65 dB	55 dB

Industrial zone	70 dB	70 dB

17.5 Control Measures

- Source control: acoustic treatment to machine surfaces; change machine design; proper lubrication.
- Transmission path: insulating enclosures; soundproof rooms.
- Receptor control: alter work schedule; use earplugs; dissipation and deflection.
- Ban noise-polluting vehicles.
- Plant trees along roadsides (absorb noise).
- Enforce noise pollution control act.
- Educate people about noise pollution.

18. SOLID, HAZARDOUS & E-WASTE MANAGEMENT

18.1 Types of Solid Wastes

Type	Sub-types / Examples
Urban/Municipal	Domestic (food waste, cloth, paper, glass, polythene); Commercial (packaging, cans, bottles, rubber); Construction (wood, concrete, cement, PoP); Biomedical (anatomical, infectious wastes)
Industrial	Nuclear power plants (radioactive waste); Thermal power plants (flyash, hot water); Chemical industries (toxic chemicals, acids); Other (packing material, oil, dyes, cement, rubber)
Hazardous	Toxic (acute/chronic); Reactive (react with air/water/heat — e.g., gun powder); Corrosive (destroy tissues — e.g., acids, bases); Infectious (used bandages, hypodermic needles)

18.2 Waste Management Hierarchy (5R)

Priority order: Reduce → Reuse → Recycle → Recover → Dispose

Reduce + Reuse + Recycle = Waste Prevention. Final step = Dispose (landfill/incineration/composting).

> 🧠 ***MNEMONIC: 'Raju Runs Races Remarkably Devotedly' = Reduce, Reuse, Recycle, Recover, Dispose***

18.3 Methods of Disposal

- Landfill: Spread waste on land → compacts over years → covered by soil.
- Incineration: Waste is burned and reduced.
- Composting: Organic waste decomposed → useful fertilizer for plants.

18.4 E-Waste

Electronic waste = old, end-of-life electronic appliances (computers, TVs, mobile phones, DVD players, mp3 players, servers, etc.) disposed by users. Hazardous constituents: lead, cadmium, mercury, plastics.

Disposal methods: Landfill, Incineration, Reuse (remove spare parts for other applications), Recycle (recover ingredients → send to manufacturers).

19. ENVIRONMENTAL PROTECTION ACTS IN INDIA

19.1 Overview

Environment Protection Act (EPA) — introduced to make provisions for controlling pollution. In 1980, Government of India established an independent department. Ministry of Environment and Forest oversees Central and State Pollution Control Boards.

19.2 Major Environmental Protection Acts

Act	Year	Objectives & Key Points
Air (Prevention and Control of Pollution) Act	1981	1. Prevent and control air pollution. 2. Maintain air quality. 3. Establish pollution control boards at various levels to check air quality. Air pollutant = solid, liquid or gaseous substance injurious to living beings.
Water (Prevention and Control of Pollution) Act	1974	1. Prevent and control water pollution. 2. Maintain water quality. 3. Establish pollution control boards to check and control water quality. Water pollution = changes in physical, chemical, biological properties injurious to ecological system / public health / domestic or agricultural use.
Wildlife Protection Act	1972	1. Maintain essential ecological processes and life supporting systems. 2. Preserve biodiversity. 3. Protect wildlife. Wildlife = any animal, aquatic/land/water, and vegetation that is the natural home of any wild animal.
Forest Conservation Act	1980	1. Protection and conservation of forests. 2. Ensure proper use of forest produce. Forest produce: timber, charcoal, oil, resin, lac, gum, seeds, silk, sandalwood, rocks, pharmaceutical plants.
Environment (Protection) Act	1986	Comprehensive umbrella legislation. Confers powers on Central Government to take measures for protecting environment quality, coordinate state actions, plan nationwide programmes, lay down standards for pollutant discharge, establish environmental laboratories, restrict industrial areas, prescribe safeguards for hazardous substances. Stringent penalties for violation. Civil court jurisdiction barred.
Biological Diversity Act	2002	Implements CBD 1992. Establishes National Biodiversity Authority (Chennai, est. 2003). Aims at conservation of biological resources and associated knowledge; facilitates access sustainably. Manages permissions for foreigners/NRIs to obtain biological resources (Section 3), transfer research results (Section 4).

🧠 MNEMONIC: 'AWWFE-B' = Air (81), Water (74), Wildlife (72), Forest (80), Environment (86), Biological Diversity (2002)

⚡ EXAM TIP: Remember years: Wildlife & Forest Act are from 1972/1980 (pre-independence era mindset). Water Act (1974) came before Air Act (1981). EPA (1986) is the umbrella act.

19.3 Salient Features of EPA (1986)

- Central Government empowered to: take all measures for protecting environment; plan nationwide programmes; lay down standards for pollutant discharge; establish environmental laboratories; restrict industrial areas.
- Persons can complain to courts regarding violations — after 60-day notice to prescribed authorities.
- Person in charge of a place MUST inform authorities about any accidental pollutant discharge exceeding standards.
- Expenses of remedial measures recoverable with interest from the polluter.
- Stringent penalties for violation.
- Jurisdiction of civil courts is barred.

19.4 OH&S Management System (OHSMS)

Occupational Health and Safety Management System = fundamental part of an organisation's risk management strategy.

Definition (ILO-OSH 2001): 'A set of interrelated or interacting elements to establish OSH policy and objectives and to achieve those objectives.'

Benefits: Implements OHS management; identifies OHS performance parameters; identifies significant hazards and risks; protects workforce; ensures legal compliance; facilitates continual improvement.

20. ENVIRONMENTAL IMPACT ASSESSMENT (EIA)

20.1 Definition

EIA = procedure for ensuring that potential environmental effects of any new development or project are considered BEFORE it is allowed to proceed. Also called Environmental Assessment (EA).

20.2 Stages of EIA

Stage	Description
1. Project Description	Clear description of the project and its location. Sufficient for planning authority to judge if EIA is needed.
2. Screening	Determining for a particular project whether an environmental assessment is necessary.
3. Scoping	Directing EIA towards aspects of specific importance; identifies aspects requiring detailed study; develops an EIA programme.
4. Baseline Studies	Identification of significant environmental impacts to be assessed. Follows scoping. Covers project details, project environs, social dimension.
5. Impact Prediction	Assessing potential environmental effects of aspects identified in scoping and baseline studies.
6. Mitigation Assessment	Consideration of measures to alleviate or minimize environmental effects.
7. Environmental Statement (ES)	Publicly available document submitted to planning authority — the key output. Responsibility of the developer.
8. Environmental Monitoring	Monitoring environmental effects after project approval. Essential foundation for environmental regulation and auditing.

> 🧠 ***MNEMONIC: 'Psst! Silly Bears Ignore Painful Monkey Stuff!' = Project desc, Screening, Scoping, Baseline, Impact, Prediction, Mitigation, Statement***

20.3 Benefits of EIA

- Provides basis for better decision making.
- Ensures potential environmental effects are fully considered.
- Allows formulation of projects within a framework of greater safeguard.
- Promotes greater interaction between developer and planning authorities.

- Provides judgmental processes administered more systematically.

QUICK REFERENCE: KEY FACTS FOR EXAM

Topic	Key Fact to Remember
Hot-spots in India	2 out of 25 global: Eastern Himalayas + Western Ghats
Hot-spot criteria	Lost >70% original habitat + rich endemism + under threat
Nitrification bacteria	Nitrobacter, Nitrosomonas (NH_3 → Nitrates)
Denitrification bacteria	Pseudomonas (Nitrates → N_2)
Primary pollutants (5)	CO, CO_2, NO_x, SO_x, Hydrocarbons, Particulate Matter
Secondary pollutants	H_2SO_4, HNO_3, Photochemical smog (ozone)
SO_2 proportion	Accounts for about 18% of all air pollution
NO_2 proportion	Accounts for about 6% of pollution
Lead in petrol	80-90% of lead in ambient air from combustion of leaded petrol
Nitrification (soil)	NH_3 → Nitrates by nitrifying bacteria (Nitrobacter)
Blue Baby Syndrome	Caused by excess nitrates in groundwater (Methemoglobinemia)
Eutrophication	Excess phosphorous → excessive algal growth in water bodies
Bio-magnification	Non-biodegradable pesticides concentrate in food chain
BMW Rules	Bio-medical Waste (Management and Handling) Rules, 1998 (under EPA 1986)
Waste hierarchy	Reduce → Reuse → Recycle → Recover → Dispose
N_2 in atmosphere	~78% of atmosphere is nitrogen
Types of biodiversity	Genetic, Species, Ecosystem
Values of biodiversity	Consumptive, Productive, Social, Ethical, Aesthetic, Option
In-situ conservation	Protection in natural habitat (national parks, sanctuaries, biosphere reserves)
Ex-situ conservation	Protection outside natural habitat (zoos, botanical gardens, gene banks)
Biosphere reserves in India	7 (by time of text; 12 total by year 2000)
National Parks in India	80
Wildlife Sanctuaries in India	420
Photosynthesis equation	CO_2 + H_2O → (chlorophyll/sunlight) → CH_2O + O_2
Biogeochemical cycles	Hydrological, Oxygen, Nitrogen, Carbon, Phosphorous
Silent zone noise limit	50 dB (day), 40 dB (night)
Industrial zone noise limit	70 dB (day and night)

— *End of Notes* —